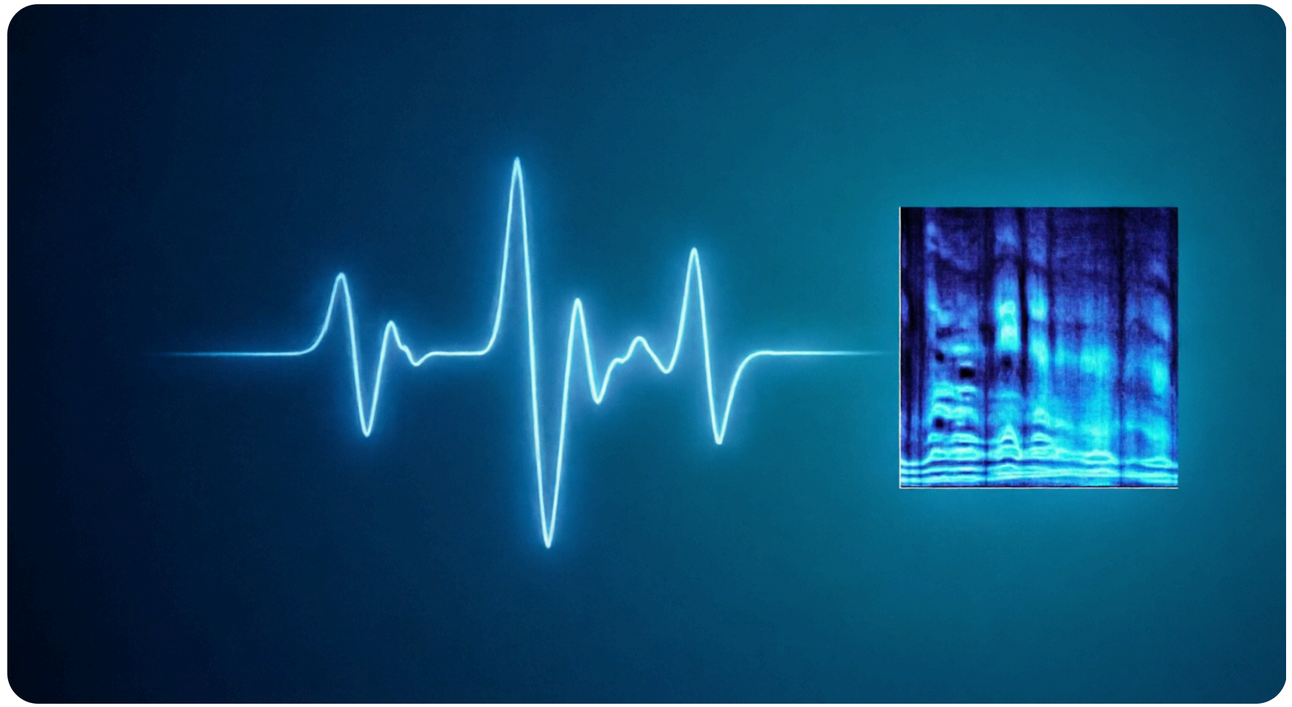
Audio Deepfake Detection

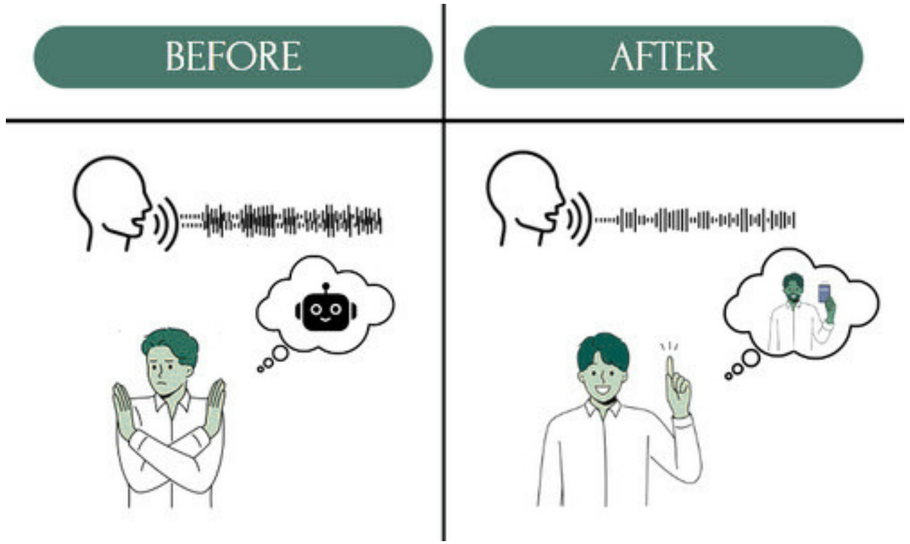


Name: Chim Lina
Final Project
Course: Deep Learning
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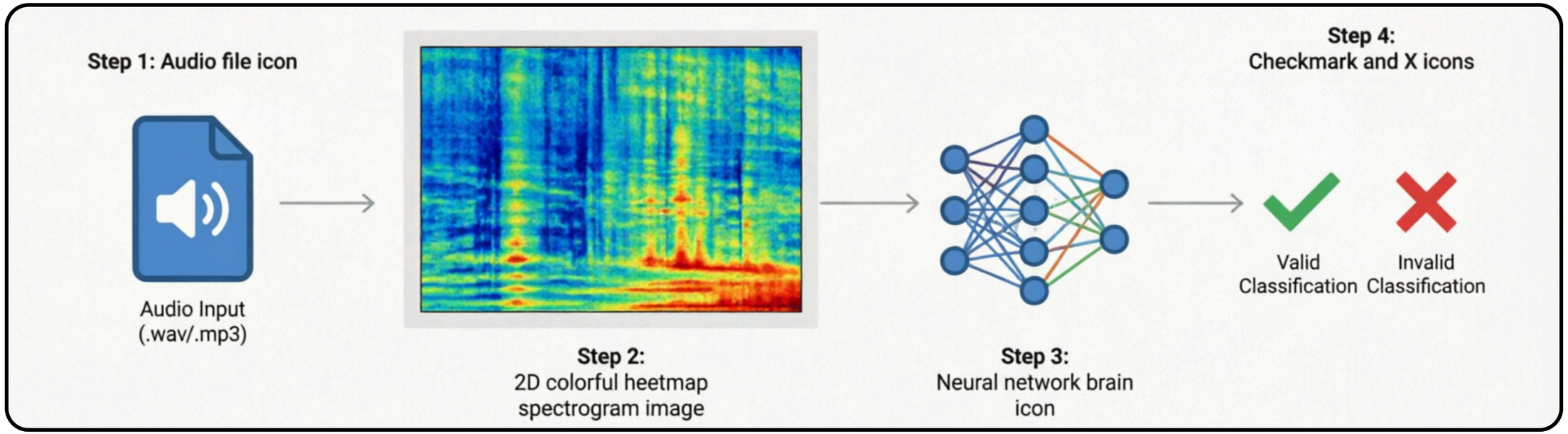
Problem Definition & Motivation

- The Threat: Rapid advancement of AI voice cloning (TTS) threatens digital trust.
- The Gap: Lack of detection tools specifically for low-resource languages like Khmer.
- The Task: Binary Audio Classification.
- Input: 3-second audio clip.

  Output: Probability of being Real (Human) vs. Fake (AI).



Dataset & Pipeline

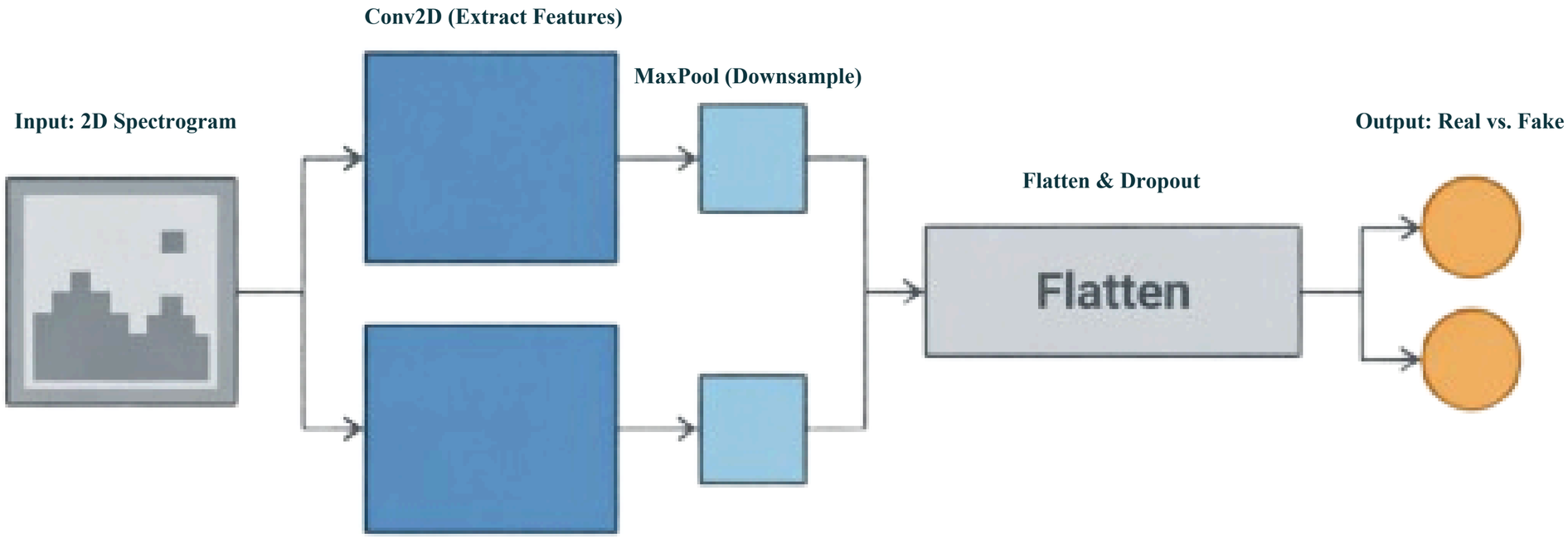


- Preprocessing Pipeline:

1. Resample to 16kHz Mono.
2. Pad/Truncate to exactly 3 seconds (48,000 samples).
3. Convert to 2D Log-Mel Spectrograms (64 mel bins).

- Dataset: ~20,109 audio files (Balanced Real/Fake).
- Data Split: Fixed 80% Train / 10% Val / 10% Test (Seed=42).

Approach 1 - Custom CNN

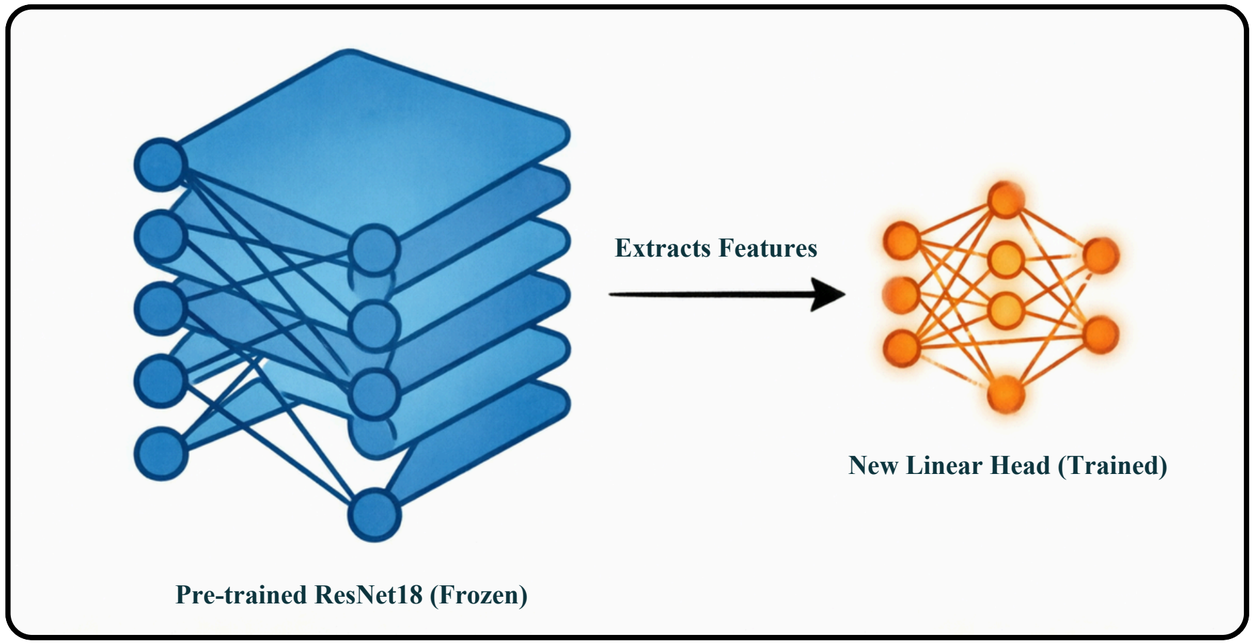


- Built from scratch (~105,000 parameters).
- Uses Locality & Weight Sharing (Session 05).

Tests if a small model can learn audio artifacts alone.

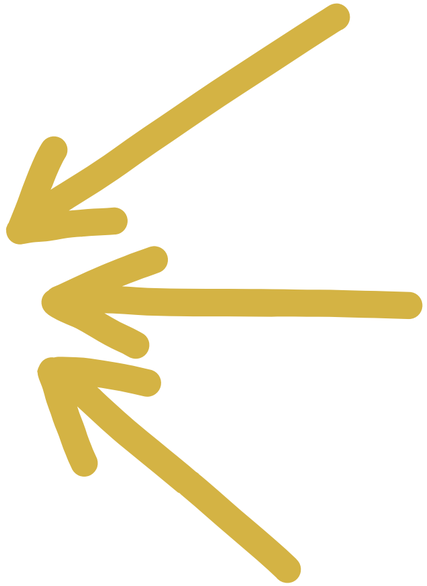
Approach 2 - ResNet18 Transfer Learning

- Pre-trained on ImageNet (~11.2 Million parameters).
- 1-channel spectrogram duplicated to 3 channels.
- Borrows pre-learned edge/texture detectors .

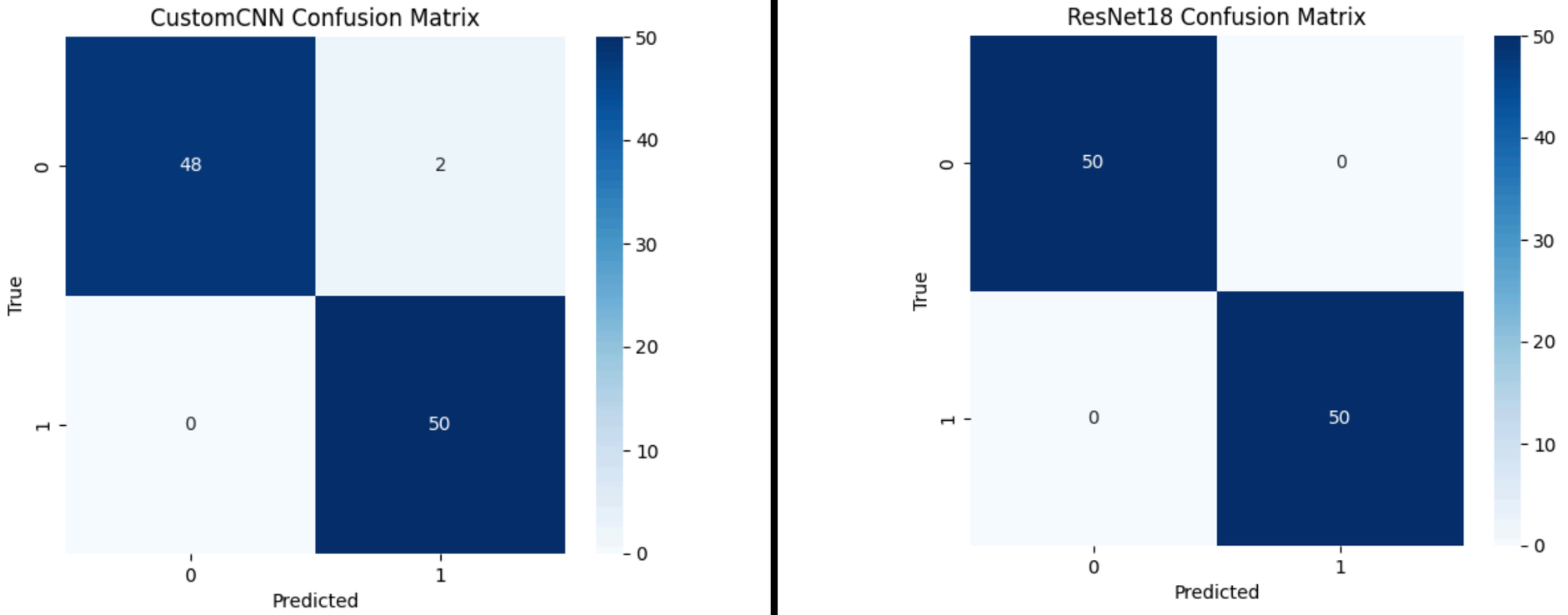


Experimental Setup

Setting	Custom CNN	ResNet18
Learning Rate	0.0001	0.0001
Batch Size	32	32
Optimizer	Adam	Adam
Epochs	2	2
Parameters	~105K	~11.2M

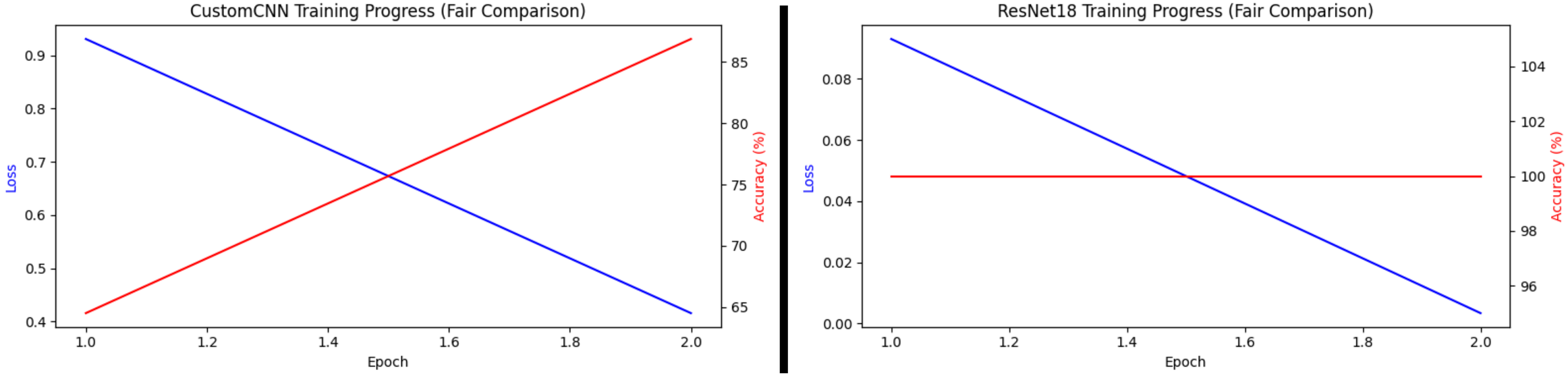


Results - Confusion Matrices



- Both models achieved 100% accuracy on the test subset.
- Zero False Positives, Zero False Negatives.

Results - Training Curves



- ResNet18 converged significantly faster.
- Custom CNN required more epochs to reach the same loss.

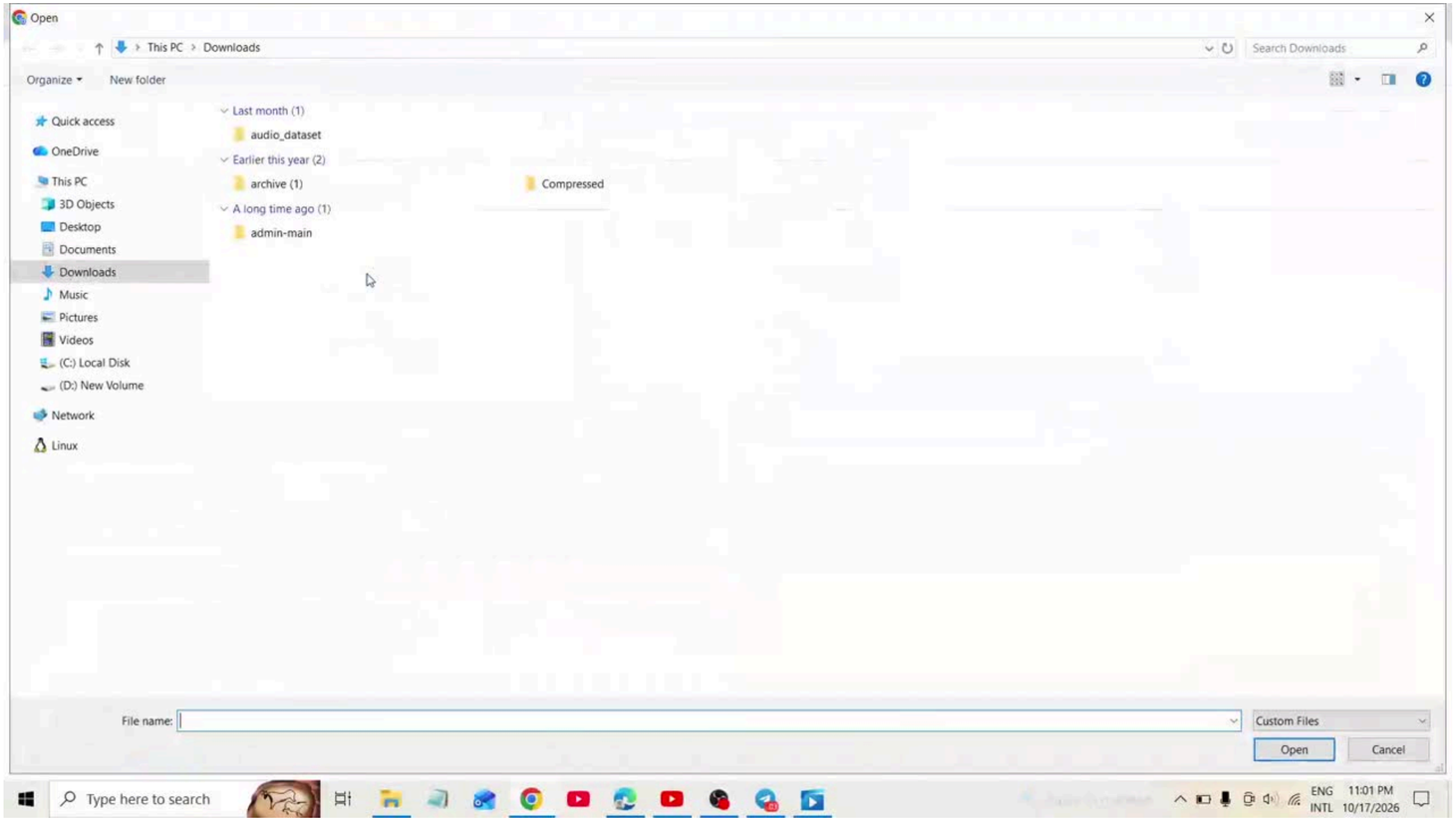
Prototyping & Web Application

- Tool: Gradio (Python).
- Features:

_ Drag-and-drop audio upload / Microphone recording.

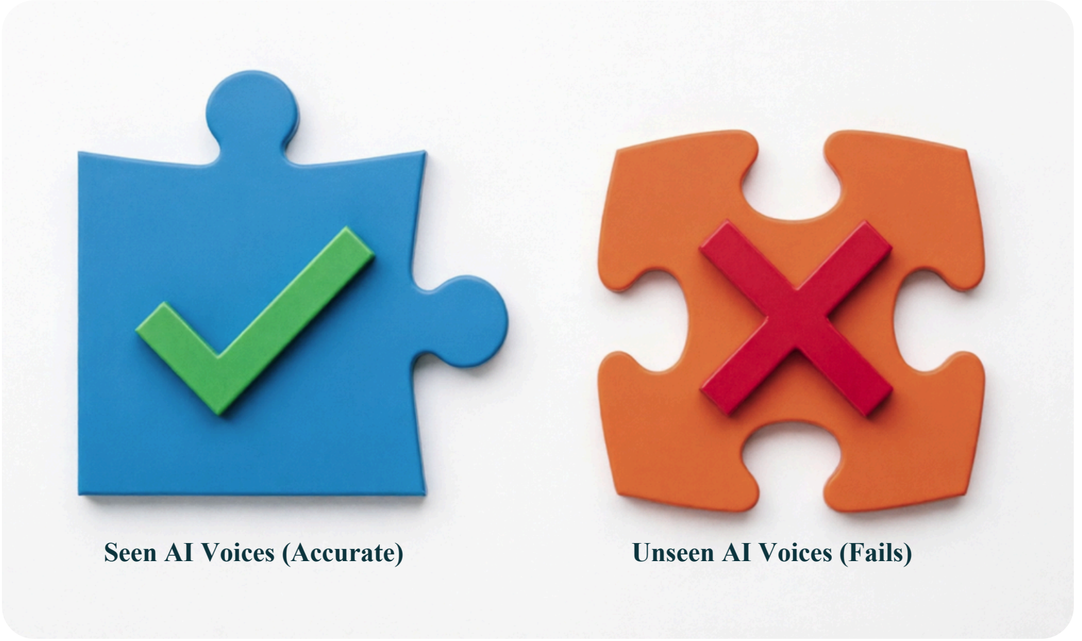
_ Real-time confidence bar chart.

_ Model ResNet18 .



Discussion & Error Analysis

- Custom CNN (Overfitting): Low capacity model memorized specific training artifacts instead of learning general "fake" features .
- ResNet18 (Generalization): Pre-trained ImageNet weights acted as a massive regularizer, forcing the model to look for fundamental structural anomalies .
- The "Domain Gap": The model struggles with unseen AI generators (ex. ElevenLabs) not present in the training distribution.
- Takeaway: Deepfake detection is an arms race; models require diverse, multi-engine datasets to generalize in the real world.



Conclusion & Future Work



- Conclusion: Transfer Learning (ResNet18) is vastly superior for spectrogram analysis on limited data.
- Future Work:
  - Add Data Augmentation
  - Train on diverse, multi-engine AI datasets to fix Domain Shift.

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Thank You!